web, interact with content, collaborate with others, and personalise their virtual spaces.

Virtual reality browsing. Janus VR enables users to browse the web in a fully immersive VR environment. Users can navigate websites, interact with content, and explore virtual spaces, all within the context of VR.

Navigation and teleportation. Users can move around virtual environments through various navigation options, including walking, flying, or teleportation. This feature allows for seamless exploration and interaction within the VR space.

Multiuser collaboration. Janus VR supports multiuser collaboration, enabling users to interact with others in real time. Users can meet and communicate with people from around the world, enhancing the social aspect of the VR experience.

Web content integration. It integrates web content seamlessly into virtual environments. Users can view websites, watch videos, access web applications, and interact with traditional web content, transforming the browsing experience into a spatial and immersive one.

Customisation and personalisation. Users can personalise their virtual spaces within Janus VR. They can customise the appearance of their virtual surroundings, create virtual objects, and manipulate the environment to suit their preferences and needs.

For developers, JanusVR provides a range of tools and features to create interactive VR experiences. With support for WebVR integration, scripting capabilities, and collaboration tools, developers can unleash their creativity.

WebVR integration. Janus VR supports WebVR, a JavaScript API that enables the creation of VR experiences on the web. Developers can build VR-enabled websites and applications that seamlessly integrate with Janus VR, expanding the possibilities of immersive web content.

Fig. 3: JanusVR render engine

Janus markup language (JML). JML is Janus VR's scripting language, specifically designed for creating interactive VR environments. Developers can use JML to define and manipulate objects, add interactivity, and create immersive experiences within Janus VR.

Collaboration tools. Janus VR provides developers with tools for creating multiuser experiences. Developers can build virtual spaces where multiple users can interact, collaborate, and communicate in real time, fostering social engagement and shared experiences.

Custom 3D models and environments. Developers can create and import custom 3D models and environments into Janus VR. This feature allows for the creation of unique and visually appealing virtual spaces, enhancing the immersive experience for users.

Interactivity and animation. Janus VR supports interactivity and animation through its scripting capabilities. Developers can use JavaScript to create dynamic and interactive elements within the VR environment, adding life and engagement to virtual experiences.

Cross-platform compatibility. The platform is compatible with various VR platforms, including headsets such as Oculus Rift, HTC Vive, and Windows Mixed Reality. This cross-platform compatibility allows developers to reach a broader audience and ensures that their VR experiences can be accessed by users on different devices.

Mozilla Hubs

Mozilla Hubs (*https://hubs.mozilla.com*) is an innovative social virtual reality (VR) platform developed by Mozilla. It enables users to connect and interact with each other in virtual environments, fostering collaboration, communication, and creativity.

At its core, Mozilla Hubs is a web-based VR platform, utilising WebVR and WebXR technologies. It allows users to access and experience virtual environments directly through web browsers, without the need for additional software installations or downloads. The web-based nature of Mozilla Hubs makes it easily accessible to a wide range of users, as it can be accessed on various devices, including VR headsets, desktop computers, and even mobile devices.

One of the key features of Mozilla Hubs is its ability to create and customise virtual spaces. Users can design and build their own immersive environments, complete with interactive objects, spatial audio, and ambient lighting. The platform supports 3D model import, enabling users to integrate their own custom objects and elements into their virtual spaces. This customisation aspect allows for endless creativity and personalisation, making each virtual environment unique and tailored to specific needs.

Mozilla Hubs facilitates real-time communication and collaboration between users. Users can interact and engage with each other using voice